CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 11

Name (Print):_____ Section:_____

1. **Understanding `this` in Java:**

   - `this.field` → access current object's data
   - `this.method()` → call current object's method
   - `this(...)` → call another constructor (same class)

   For each line below, choose what `this` is doing:
   A. Access the current object's data
   B. Call the current object's method
   C. Call another constructor from the same class

```java
class Drink {
    String name;  int calories;
    public Drink(String name, int calories) {
        this.name = name;
        this.calories = calories;
    }
    public Drink(String name) {
        this(name, 0);
    }
    public void printInfo() {
        this.display();
    }
    public void display() {
        System.out.println(this.name + " " + this.calories);
    }}
```

   (a) `this(name, 0);` → _____
   (b) `this.name = name;` → _____
   (c) `this.display();` → _____
   (d) `this.name` → _____

2. **What happens if we move `this(...)` to the second line?**

```java
public Drink(String name, int calories) {
    System.out.println("Creating drink...");
    this(name, calories, false);
}
```

   (a) OK

(b) Runtime error
(c) Compile-time error

3. **Relationships**.Select from the following: **IS-A**, **HAS-A**, **DEPENDS-ON**

   (a) `class Car implements Drivable`: _____
   (b) `class Car { private Engine engine; }`: _____
   (c) `class Vet { void examine(Dog d) {...}}`: _____

4. **Definition (Inheritance)**
   Inheritance is a mechanism when a _____ automatically gains the fields and methods of its _____ by using the `extends` keyword.

5. **Java class hierarchy**
   What is the name of the topmost class in Java? _____

6. **What can a subclass do? (circle/select all correct)**

   (a) A subclass inherits fields and methods from its superclass, but private members are not directly accessible.
   (b) A subclass inherits constructors from its superclass.
   (c) A subclass must call (explicitly or implicitly) a superclass constructor.
   (d) A subclass can override public and protected methods (if they are not `final` or `static`).
   (e) A subclass can override private methods.

7. **Using `super`**. Fill in the correct Java code.

   (a) Call a superclass method `eat()`: _____
   (b) Call a superclass constructor `Animal(String name)`: _____

8. **Overriding: what prints?**. Write the output for each line:

```java
class Animal {
    void move() { System.out.println("Animal is moving."); }
    void eat()  { System.out.println("Animal is eating."); }
}
class Dog extends Animal {
    @Override
    void move() {System.out.println("Dog is running."); }
    void bark() { System.out.println("Dog is barking."); }
}
public class Main {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.move();  // Output 1: __________________
        d.eat();   // Output 2: __________________
        d.bark();  // Output 3: __________________
    }}
```

9. **Before You Leave**
   Write one question you still have about today's topic.